



Cardiovascular pharmacology

Intrathecal dihydroergotamine inhibits capsaicin-induced vasodilatation in the canine external carotid circulation via GR127935- and rauwolscine-sensitive receptors

Bruno A. Marichal-Cancino^a, Abimael González-Hernández^a, Guadalupe Manrique-Maldonado^a, Inna I. Ruiz-Salinas^a, Alain H. Altamirano-Espinoza^a, Antoinette MaassenVanDenBrink^b, Carlos M. Villalón^{a,*}

^a Departamento de Farmacobiología, Cinvestav-Coapa, Czada. de los Tenorios 235, Col. Granjas-Coapa, Deleg. Tlalpan, 14330 Mexico D.F., Mexico

^b Division of Vascular Medicine and Pharmacology, Department of Internal Medicine, Erasmus University Medical Center, PO Box 2040, 3000 CA Rotterdam, The Netherlands

ARTICLE INFO

Article history:

Received 28 March 2012

Received in revised form

22 June 2012

Accepted 10 July 2012

Available online 27 July 2012

Keywords:

Capsaicin

CGRP

Dihydroergotamine

External carotid artery

Migraine

ABSTRACT

It has been suggested that during a migraine attack trigeminal nerves release calcitonin gene-related peptide (CGRP), producing central nociception and vasodilatation of cranial arteries, including the extracranial branches of the external carotid artery. Since trigeminal inhibition may prevent this vasodilatation, the present study has investigated the effects of intrathecal dihydroergotamine on the external carotid vasodilatation to capsaicin, α -CGRP and acetylcholine. Anaesthetized vagosympathectomized dogs were prepared to measure blood pressure, heart rate and external carotid conductance. A catheter was inserted into the right common carotid artery for the continuous infusion of phenylephrine (to restore the carotid vascular tone), whereas the corresponding thyroid artery was cannulated for one-min intracarotid infusions of capsaicin, α -CGRP and acetylcholine (which dose-dependently increased the external carotid conductance). Another cannula was inserted intrathecally (C₁–C₃) for the administration of dihydroergotamine, the α_2 -adrenoceptor antagonist rauwolscine or the serotonin 5-HT_{1B/1D} receptor antagonist GR127935 (N-[4-methoxy-3-(4-methyl-1-piperazinyl) phenyl]-2'-methyl-4'-(5-methyl-1,2,4-oxadiazol-3-yl)[1,1-biphenyl]-4-carboxamide hydrochloride monohydrate). Intrathecal dihydroergotamine (10, 31 and 100 μ g) inhibited the vasodilatation to capsaicin, but not that to α -CGRP or acetylcholine. This inhibition was: (i) unaffected by 10 μ g GR127935 or 100 μ g rauwolscine, but abolished by 31 μ g GR127935 or 310 μ g rauwolscine at 10 μ g dihydroergotamine; and (ii) abolished by the combination 10 μ g GR127935 + 100 μ g rauwolscine at 100 μ g dihydroergotamine. Thus, intrathecal (C₁–C₃) dihydroergotamine seems to inhibit the external carotid vasodilatation to capsaicin by spinal activation of serotonin 5-HT_{1B/1D} (probably 5-HT_{1B}) receptors and α_2 (probably $\alpha_{2A/2C}$) adrenoceptors.

© 2012 Elsevier B.V. All rights reserved.

1. Introduction

Migraine has classically represented an ancient ailment, the pathophysiology of which was, until recently, debatable (Dodick, 2008). Indeed, our understanding of migraine has evolved slowly from a malady of supernatural causes (Villalón et al., 2003) to a disorder of vascular (Graham and Wolff, 1938), neurogenic (Moskowitz et al., 1979) or neurovascular (Villalón and Olesen, 2009) origin. Although it has even been claimed that its vascular

changes may be an epiphenomenon (e.g., Akerman et al., 2011), migraine is currently considered a neurovascular disorder involving: (i) activation of the trigemino-vascular system (Olesen et al., 2009), with the primary dysfunction located in brainstem centres regulating vascular tone and pain sensation (Link et al., 2008); (ii) release of vasoactive neuropeptides from trigeminal sensory nerves including calcitonin gene-related peptide (CGRP), which plays a role in migraine pathophysiology (Villalón and Olesen, 2009); and (iii) vasodilatation of intracranial arteries and of the extracranial branches of the external carotid artery, which are a source of pain in migraine (Olesen et al., 2009; Shevel, 2011a, 2011b). Thus, trigeminal inhibition may prevent this vasodilatation and abort migraine.

CGRP release from perivascular sensory (including trigeminal) nerves can be induced by capsaicin (Dux et al., 2003), which

* Corresponding author. Tel.: +52 55 5483 2854;

fax: +52 55 5483 2863(Int.).

E-mail address: cvillalon@cinvestav.mx (C.M. Villalón).

¹ <http://www.farmacobiologia.cinvestav.mx/PersonalAcadémico/DrCarlosM.VillalónHerrera.aspx>

Table 1
Binding affinity constants (pK_i) of dihydroergotamine, rauwolscine or GR127935 for the different subtypes of cloned serotonin 5-HT₁ receptors and α_2 -adrenoceptors.

Compound	5-HT _{1A}	5-HT _{1B}	5-HT _{1D}	5-HT _{1E}	5-HT _{1F}	α_2	α_{2A}	α_{2B}	α_{2C}
Dihydroergotamine	9.3 ^a	(r) 7.5 ^a	8.6 ^a	6.2 ^a	6.9 ^a	(r) 8.0 ^a	8.7 ^a	8.0 ^a	9.0 ^a
Rauwolscine	–	6.5 ^b	7.8 ^b	5.5 ^b	–	–	8.9 ^c	8.9 ^c	9.3 ^c
GR127935	–	9.0 ^d	8.6 ^d	5.4 ^d	6.4 ^d	< 6.0 ^e	–	–	–

All data are given as pK_i values at human recombinant receptors, except when stated otherwise: rodent (r) receptors.

^a Data taken from: Leysen et al. (1996),

^b Data taken from: Uhlén et al. (1994),

^c Data taken from: Jasper et al. (1998),

^d Data taken from: Price et al. (1997),

^e Data taken from: Pauwels (1996).

activates the transient receptor potential vanilloid type 1 (TRPV1) cation channel (Caterina et al., 1997; Kim et al., 2012). Moreover, capsaicin-induced vasodilatation in the canine external carotid circulation is inhibited by spinal activation of either serotonin 5-HT_{1B} receptors (by intrathecal sumatriptan; Muñoz-Islas et al., 2009) or $\alpha_{2A/2C}$ -adrenoceptors (by intrathecal azepexole [B-HT 933]; Villalón et al., 2012). Since dihydroergotamine is effective in the acute treatment of migraine (Saper and Silberstein, 2006) and displays affinity for serotonin 5-HT₁ receptors and α_2 -adrenoceptors (Table 1), it is tempting to suggest that the above central mechanisms could be, at least in part, involved in its antimigraine action. Indeed, the antimigraine action of dihydroergotamine may involve: (i) external carotid (extracranial) vasoconstriction via serotonin 5-HT_{1B} receptors and $\alpha_{2A/2C}$ -adrenoceptors (Villalón et al., 2004); and (ii) inhibition of both trigeminal pain transmission (Goadsby and Gundlach, 1991; Kayser et al., 2002) and neurogenic inflammation (Saito et al., 1988). Hence, an experimental approach that would allow us to investigate the exclusive role of central (spinal) mechanisms in the effects of dihydroergotamine is the analysis of its effects produced by intrathecal administration on capsaicin-induced vasodilatation in the canine external carotid circulation. On this basis, the objective of the present study was to investigate in vagosympathectomized dogs: (i) whether direct intrathecal (C₁–C₃) administration of dihydroergotamine results in inhibition of the external carotid vasodilator responses to intracarotid capsaicin, α -CGRP and acetylcholine; and (ii) the role of spinal serotonin 5-HT_{1B/1D} receptors and α_2 -adrenoceptors in the above inhibition to dihydroergotamine by analyzing the effects to intrathecal (C₁–C₃) administration of the antagonists GR127935 (5-HT_{1B/1D}) or rauwolscine ($\alpha_{2A/2B/2C}$) (Table 1). This is a neurovascular model of migraine that has been useful for the screening of compounds with antimigraine activity (Gupta and Villalón, 2010; Saxena and de Vlaam-Schluter, 1974; Shevel, 2011a, 2011b; Villalón et al., 2003).

2. Materials and methods

2.1. General methods

Experiments were carried out in a total of 66 male mongrel dogs (body weight: 18–25 kg). The animals were anaesthetized with an intravenous (i.v.) bolus injection of sodium pentobarbitone (30 mg/kg), and additional amounts (1 mg/kg, i.v.) were provided when required throughout the experiment for maintaining the anaesthesia, as previously reported (Villalón et al., 1999). All dogs were intubated with an endotracheal tube and artificially respired with room air, using a Palmer ventilation pump at a rate of 20 strokes/min (stroke volume: 13–16 ml/kg) as previously established by Kleinman and Radford (1964). A catheter was placed in the right femoral artery, connected to a blood

pressure transducer (P23XL, Grass Instrument Co., Quincy, MA, U.S.A.), for measurement of arterial blood pressure. Another catheter was placed intrathecally (Yaksh et al., 2004) through the subarachnoid space in the spinal C₁–C₃ region for the administration of vehicles (physiological saline or propylene glycol 10%), dihydroergotamine, GR127935 or rauwolscine. Since the trigeminal sensory nerve projects to the trigeminal nucleus caudalis in the medulla oblongata, the intrathecal administration of these compounds at the cervical level was carried out in an attempt to facilitate their diffusion to the medulla oblongata (with their resulting effects on perivascular trigeminal sensory terminals). After administration of compounds, the intrathecal catheter was flushed with 0.3 ml of saline. Mean arterial blood pressure (MAP) was calculated from the systolic (SAP) and diastolic (DAP) arterial pressures as follows: $MAP = DAP + (SAP - DAP)/3$. Heart rate was measured with a tachograph (7P4F, Grass Instrument Co., Quincy, MA, U.S.A.) triggered from the blood pressure signal.

The right common carotid artery was dissected free and the corresponding internal carotid and occipital branches were ligated. Under these experimental conditions, the blood flow through the right common carotid artery was considered to represent that of the external carotid artery (for further considerations see Villalón et al., 1999). Thereafter, an ultrasonic flow probe (4 mm R-Series) connected to an ultrasonic T206 flow meter (Transonic Systems Inc., Ithaca, NY, U.S.A.) was placed around the right common carotid artery. Bilateral cervical vagosympathectomy was systematically performed in order to prevent possible baroreceptor reflexes induced by the intracarotid infusions of compounds. Subsequently, a catheter was introduced into the right cranial thyroid artery for the intracarotid infusions of capsaicin, α -CGRP and acetylcholine.

It is to be noted that after bilateral vagosympathectomy the carotid arterioles are dilated under our experimental conditions (Villalón et al., 1995). Since the main objective of our study is to investigate the responses to vasodilator agents (such as capsaicin, α -CGRP and acetylcholine) on the external carotid circulation, we had to produce a selective carotid precontraction with an intracarotid continuous infusion of phenylephrine (an α_1 -adrenoceptor agonist); this procedure allowed us to enhance the vasodilator responses to capsaicin, α -CGRP and acetylcholine, as previously reported (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006, 2009). For this purpose, a 0.5 mm (external diameter) needle, connected to a suitable catheter, was inserted into the right common carotid artery for the continuous infusion of phenylephrine by another motor driven syringe. By using WPI model sp100i pumps (World Precision Instruments Inc., Sarasota, FL, U.S.A.), the vasodilator responses to capsaicin, α -CGRP and acetylcholine during the infusion of phenylephrine were compared with those observed after intrathecal administration of dihydroergotamine, antagonists or equivalent volumes of vehicle (for further details, see the protocols described below).

Arterial blood pressure, heart rate and external carotid blood flow were recorded simultaneously by a model 7D polygraph (Grass Instrument Co., Quincy, MA, U.S.A.). The body temperature of the animals was maintained between 37 and 38 °C.

2.2. Experimental protocols

After a stable haemodynamic condition for 60 min, baseline values of mean blood pressure, heart rate and external carotid blood flow were determined. Then, all animals received an intracarotid continuous infusion of phenylephrine ranging from 1.5 to 6.8 µg/min (most likely due to biological variance as mongrel dogs of 18–25 kg were used; see above) to get a sustained carotid vasoconstriction (a 40–50% decrease in external carotid blood flow) (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006, 2009; Villalón et al., 2012). After 20 min, all the treatments were given without interrupting the infusion of phenylephrine. Then, the 66 animals were divided into five main groups ($n=18, 12, 12, 18$ and 6, respectively) that received the following treatments.

- The first group ($n=18$) received consecutive intracarotid infusions (1 ml/min, for 1 min) of capsaicin (10, 18, 31 and 56 µg/min) (control curve). Then, this group was subdivided into three subgroups ($n=6$ each) that received intrathecal bolus injections (0.5 ml) of, respectively: (a) vehicle (propylene glycol 10%; three times); (b) dihydroergotamine (3.1, 10 and 31 µg given cumulatively); and (c) dihydroergotamine (10, 31 and 100 µg given cumulatively). Ten min after intrathecal administration of each dose of dihydroergotamine or its equivalent volume of propylene glycol 10% (0.5 ml), the dose–response curves to capsaicin were elicited again (completed 30 min after starting the infusion of the first dose of capsaicin). The next doses of dihydroergotamine or equivalent volumes of vehicle were given when the administration scheme of capsaicin had been finished (i.e., when the external carotid conductance returned to baseline values after the last dose of capsaicin).
- The second group ($n=12$) received consecutive intracarotid infusions (1 ml/min, for 1 min) of capsaicin (10, 18, 31 and 56 µg/min), α -CGRP (0.1, 0.31, 1 and 3.1 µg/min) and acetylcholine (0.01, 0.031, 0.1 and 0.31 µg/min). Subsequently, this group was subdivided into two subgroups that received intrathecal bolus injections (0.5 ml) of: (a) vehicle (propylene glycol 10%; $n=6$); or (b) 10 µg dihydroergotamine ($n=6$). Ten min later, the responses to the above infusions of capsaicin, α -CGRP and acetylcholine were elicited again.
- The third group ($n=12$), which received the intracarotid infusions of capsaicin as described above, was subdivided into three subgroups ($n=4$ each) that received intrathecal bolus injections (0.5 ml) of, respectively: (a) physiological saline (vehicle); (b) GR127935 (31 µg); or (c) rauwolscline (310 µg). Ten min later, the responses to the above infusions of capsaicin were elicited again.
- The fourth group ($n=18$) received the intracarotid infusions of capsaicin as described above. After 10 min, this group was given an intrathecal bolus injection (0.5 ml) of 10 µg dihydroergotamine and, 10 min later, was subdivided into three subgroups ($n=6$ each) that received intrathecal bolus injections (0.5 ml each) of, respectively: (a) vehicle (physiological saline, given twice); (b) GR127935 (10 and 31 µg, given cumulatively); and (c) rauwolscline (100 and 310 µg, given cumulatively). After 10 min, the responses to the above infusions of capsaicin were elicited again after each dose of vehicle or antagonist.
- The fifth group ($n=6$) was given the intracarotid infusions of capsaicin (control curve) as described above. Ten min later,

this group received successive treatment (allowing 10 min after each treatment) with: (i) an intrathecal bolus injection (0.5 ml) of 100 µg dihydroergotamine; (ii) an intrathecal bolus injection (1 ml) of physiological saline followed by the responses to the above infusions of capsaicin; and (iii) an intrathecal bolus injection of the combination GR127935 (10 µg in 0.5 ml)+rauwolescline (100 µg in 0.5 ml) followed, again, by the responses to the above infusions of capsaicin. It is noteworthy that the experiments of this group were performed in order to investigate the possibility of a simultaneous activation of serotonin 5-HT_{1B/1D} receptors and α_2 -adrenoceptors, since 100 µg dihydroergotamine induced a more evident inhibition of the vasodilator responses to capsaicin. Admittedly, such a dose may have reached much higher spinal concentrations of dihydroergotamine than those elicited by therapeutic antimigraine treatment (probably better reflected by 10 µg dihydroergotamine).

2.3. Other procedures applying to all protocols

Each dose of capsaicin, α -CGRP and acetylcholine was in a solution administered intracarotidly at a rate of 1 ml/min during 1 min. The dose-intervals between the different doses of these compounds ranged between 5 min (acetylcholine) and 15 min (capsaicin and α -CGRP), as in each case we waited until the external carotid blood flow had returned completely to baseline values. The doses of these compounds were selected on the basis of results previously published (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006, 2009; Villalón et al., 2012), in which dose-dependent increases in external carotid blood flow were elicited with no changes in blood pressure or heart rate.

The correctness of the intrathecal administration of compounds was validated at the end of all experiments (after sodium pentobarbitone overdose) by injecting an intrathecal bolus of 0.5 ml methylene blue 1%, as reported by Muñoz-Islas et al. (2009). Then, the spinal cord was dissected by laminectomy at the level of the C₁–C₄ intervertebral discs to look for the catheter tip and dye distribution. The dogs showing the catheter tip positioned at sites other than the C₁–C₃ spinal cord were not considered for data analysis.

Based on previous findings obtained in mongrel dogs by our group using identical protocols (Muñoz-Islas et al., 2009; Villalón et al., 2012), the Ethical Committee of our Institution (CICUAL) approved all the experimental protocols of this study and the number of dogs used for each group/subgroup (see Section 2.2 above). CICUAL also followed the regulations established by the Mexican Official Norm (NOM-062-1999), in accordance with the NIH Guide for the Care and Use of Laboratory Animals.

2.4. Drugs

Apart from the anaesthetic (sodium pentobarbitone), the compounds used in the present study (obtained from the sources indicated) were: L-phenylephrine hydrochloride, capsaicin, rat α -CGRP, acetylcholine chloride, rauwolescline (Sigma Chemical Co., St. Louis, MO, U.S.A.); dihydroergotamine (gift from Novartis, Basel, Switzerland); and GR127935 (N-[4-methoxy-3-(4-methyl-1-piperazinyl)phenyl]-2'-methyl-4'-(5-methyl-1,2,4-oxadiazol-3-yl) [1,1-biphenyl]-4-carboxamide hydrochloride monohydrate) (gift from GlaxoSmithKline, Stevenage, Hertfordshire, U.K.). All compounds were dissolved in physiological saline, except dihydroergotamine (dissolved in propylene glycol 10%). These vehicles had no effect, when given intrathecally, on the external carotid blood flow, arterial blood pressure or heart rate (data not

shown). The doses of the antagonists refer to their respective salts, whilst those of the agonists refer to their free base.

2.5. Data presentation and statistical evaluation

All data are presented as mean $\pm$ S.E.M. The external carotid conductance was calculated by dividing blood flow by mean blood pressure and multiplied by hundred. The peak changes in external carotid conductance were expressed as percent change from baseline. The difference between the variables within one group (or * different groups) of animals was compared by using a two-way repeated measures analysis of variance (or, correspondingly, * a two-way analysis of variance) (both randomized block design) followed by the Student–Newman–Keuls' test. Statistical significance was accepted at $P < 0.05$ (two-tailed).

3. Results

3.1. Systemic and carotid haemodynamic effects

Baseline values of mean blood pressure, heart rate and external carotid conductance in the 66 dogs were: 125 ± 4 mm Hg, 167 ± 1 beats/min and 176 ± 8 ml/min mm Hg. Twenty min after starting the phenylephrine infusion, a significant (and sustained) decrease in external carotid conductance (roughly 50%) was induced without significant changes in mean blood pressure or heart rate ($n = 66$; see Table 2). This procedure was systematically performed to enhance the vasodilator responses to capsaicin, α -CGRP and

acetylcholine, as previously reported (Jiménez-Mena et al., 2006; Muñoz-Islas et al., 2006, 2009; Villalón et al., 2012). Under these experimental conditions, the above variables were not significantly modified in the animals that received intrathecal bolus injections of: (i) physiological saline (0.5 ml; $n = 4$); (ii) propylene glycol 10% (0.5 ml; three times; $n = 6$); (iii) dihydroergotamine (3.1, 10 and 31 μ g; $n = 6$); (iv) dihydroergotamine (10, 31 and 100 μ g; $n = 6$); (v) GR127935 (31 μ g; $n = 4$); or (vi) rauwolscline (310 μ g; $n = 4$) (see Table 3).

3.2. Effect of increasing doses of intrathecal dihydroergotamine on the external carotid vasodilator responses to capsaicin

The 1-min intracarotid infusions of capsaicin induced dose-dependent increases in external carotid conductance (Fig. 1A; control responses). These vasodilator responses to capsaicin were: (i) unchanged after intrathecal vehicle (0.5 ml propylene glycol 10% given three times; see Fig. 1A); and (ii) dose-dependently inhibited after 10 and 100 μ g dihydroergotamine given intrathecally, with 31 μ g producing the same inhibition as that by 10 μ g (see Fig. 1B and C). In no case were mean blood pressure and heart rate significantly modified (Table 2). The control responses to capsaicin in the groups treated with propylene glycol 10% (Fig. 1A) or dihydroergotamine (Fig. 1B and C) did not significantly differ (two-way ANOVA; $P > 0.05$). On the basis of the above results, 10 μ g dihydroergotamine was chosen for further pharmacological analysis since: (i) this was the lowest dose that significantly inhibited the vasodilatation to capsaicin (see Fig. 1); (ii) the central levels of dihydroergotamine are rather low in

Table 2

Values of mean arterial blood pressure (MAP; mm Hg), heart rate (HR; beats/min) and external carotid conductance (ECC; ml/min mm Hg) before and 20 min after the continuous infusion of phenylephrine (1.5–6.8 μ g/min), and in three different groups ($n = 6$ each) before and after: (i) 20 min the intracarotid infusion of phenylephrine had commenced; and (ii) 10 min after intrathecal bolus injections of propylene glycol 10% (0.5 ml; given three times); dihydroergotamine (3.1, 10 and 31 μ g given cumulatively; or 10, 31 and 100 μ g given cumulatively) during phenylephrine infusions.

Treatment	n	MAP (mm Hg)		HR (beats/min)		ECC (ml/min mm Hg)	
		Before	After	Before	After	Before	After
Phenylephrine (1.5–6.8 μ g/min)	66	125 ± 4	130 ± 5	167 ± 1	157 ± 4	176 ± 8	109 ± 4^a
Phenylephrine (1.5–6.8 μ g/min)	6	137 ± 14	142 ± 17	151 ± 8	150 ± 11	174 ± 36	111 ± 10^a
1st propylene glycol 10%		142 ± 17	152 ± 18	150 ± 11	152 ± 9	111 ± 10	101 ± 16
2nd propylene glycol 10%		152 ± 18	156 ± 15	152 ± 9	150 ± 8	101 ± 16	111 ± 10
3rd propylene glycol 10%		156 ± 15	152 ± 16	150 ± 8	150 ± 9	111 ± 10	113 ± 9
Phenylephrine (1.5–6.8 μ g/min)	6	109 ± 15	111 ± 8	162 ± 8	158 ± 10	179 ± 18	95 ± 6^a
Dihydroergotamine 3.1 μ g		111 ± 8	115 ± 7	158 ± 10	151 ± 6	95 ± 6	96 ± 7
Dihydroergotamine 10 μ g		115 ± 7	116 ± 12	151 ± 6	145 ± 5	96 ± 7	102 ± 5
Dihydroergotamine 31 μ g		116 ± 12	122 ± 12	145 ± 5	147 ± 7	102 ± 5	104 ± 2
Phenylephrine (1.5–6.8 μ g/min)	6	146 ± 15	149 ± 17	158 ± 8	160 ± 11	245 ± 20	131 ± 10^a
Dihydroergotamine 10 μ g		149 ± 17	150 ± 14	160 ± 11	147 ± 9	131 ± 10	118 ± 5
Dihydroergotamine 31 μ g		150 ± 14	156 ± 12	147 ± 9	144 ± 9	118 ± 5	115 ± 8
Dihydroergotamine 100 μ g		156 ± 12	158 ± 16	144 ± 9	145 ± 11	115 ± 8	117 ± 12

^a $P < 0.05$, after vs. before from the corresponding baseline value. All values are expressed as mean $\pm$ S.E.M.

Table 3

Values of mean arterial blood pressure (MAP; mm Hg), heart rate (HR; beats/min) and external carotid conductance (ECC; ml/min mm Hg) before and 10 min after the intrathecal bolus injections of propylene glycol (0.5 ml), dihydroergotamine (10 and 100 μ g), physiological saline (0.5 ml), GR127935 (31 μ g) and rauwolscline (310 μ g). Phenylephrine was infused exactly as described in Table 2.

Treatment	n	MAP (mm Hg)		HR (beats/min)		ECC (ml/min mm Hg)	
		Before	After	Before	After	Before	After
Propylene glycol 10%	12	151 ± 9	154 ± 10	151 ± 6	146 ± 7	124 ± 10	116 ± 5
Dihydroergotamine (10 μ g)	12	131 ± 13	136 ± 11	159 ± 6	153 ± 5	112 ± 10	98 ± 8
Dihydroergotamine (100 μ g)	6	156 ± 12	158 ± 16	144 ± 9	145 ± 11	115 ± 8	117 ± 12
Physiological saline (0.5 ml)	4	146 ± 14	146 ± 16	145 ± 10	141 ± 15	122 ± 13	115 ± 11
GR127935 (31 μ g)	4	126 ± 9	122 ± 8	155 ± 2	154 ± 2	82 ± 11	75 ± 6
Rauwolscline (310 μ g)	4	147 ± 8	151 ± 10	180 ± 1	182 ± 4	96 ± 9	90 ± 7

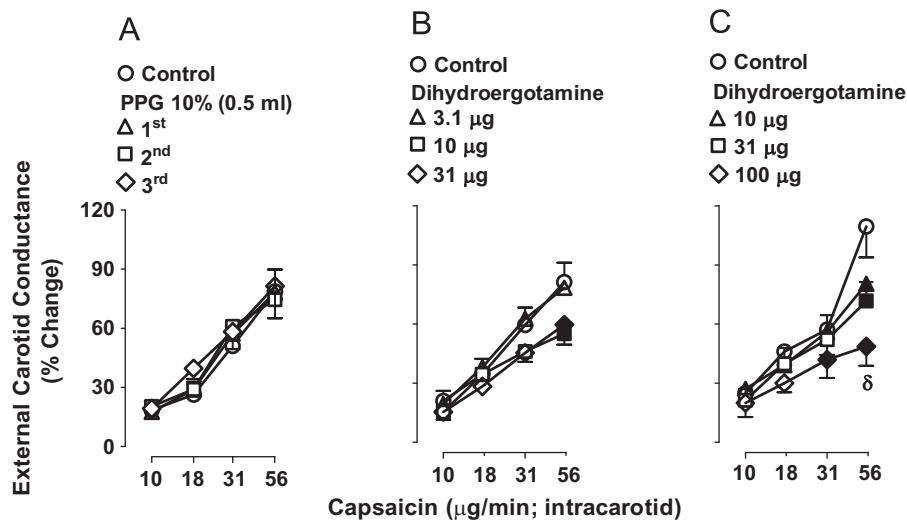


Fig. 1. Canine external carotid vasodilator responses produced by 1-min intracarotid infusions of capsaicin before (control responses) and after intrathecal bolus injections of: (A) 0.5 ml propylene glycol 10% (PPG 10%; given three times; $n=6$); (B) dihydroergotamine (3.1, 10 and 31 μg given cumulatively; $n=6$); or (C) dihydroergotamine (10, 31 and 100 μg given cumulatively; $n=6$). For the sake of clarity, empty symbols depict either control responses ($\circ$) or non-significant responses ($\triangle$ $\square$ $\diamond$) vs. control. Solid symbols ($\blacktriangle$ $\blacksquare$ $\blacklozenge$) represent significantly different responses ($P < 0.05$) vs. control; δ , $P < 0.05$ when comparing the inhibition by 31 μg dihydroergotamine vs. 100 μg dihydroergotamine; Fig. 1C). Note that the control responses in (A) did not significantly differ from those in ((B) and (C)) ($P > 0.05$).

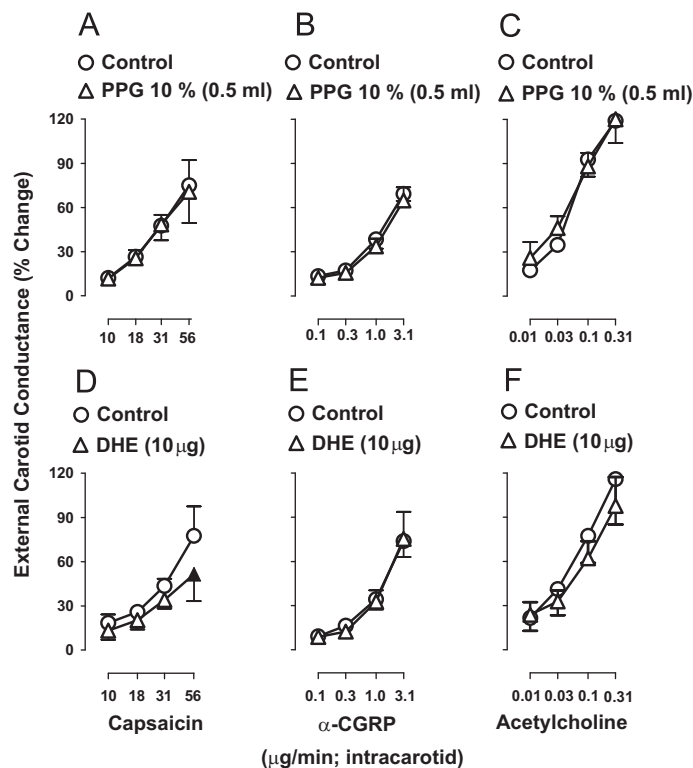


Fig. 2. Canine external carotid vasodilator responses produced by 1-min intracarotid infusions of capsaicin, α -CGRP or acetylcholine before (control responses) and after an intrathecal bolus injection of either ($n=6$ each): (i) 0.5 ml propylene glycol 10% (PPG 10%; (A)–(C)); or (ii) 10 μg dihydroergotamine (DHE; (D)–(F)). For the sake of clarity, empty symbols depict either control responses ($\circ$) or non-significant responses ($\triangle$) vs. control. Solid symbols ($\blacktriangle$) represent significantly different responses ($P < 0.05$) vs. control.

antimigraine therapy (Saper and Silberstein, 2006) because it exhibits a moderate blood-brain barrier permeability and brain-stem distribution when given intranasally or intravenously (Wang et al., 1998); and (iii) this intrathecal low dose may closely approximate the therapeutic reality (with all due considerations, including species differences).

3.3. Effect of intrathecal propylene glycol 10% or 10 μg dihydroergotamine on the external carotid vasodilator responses to capsaicin, α -CGRP and acetylcholine

As shown in Fig. 2, the intracarotid infusions of capsaicin, α -CGRP and acetylcholine induced dose-dependent increases in

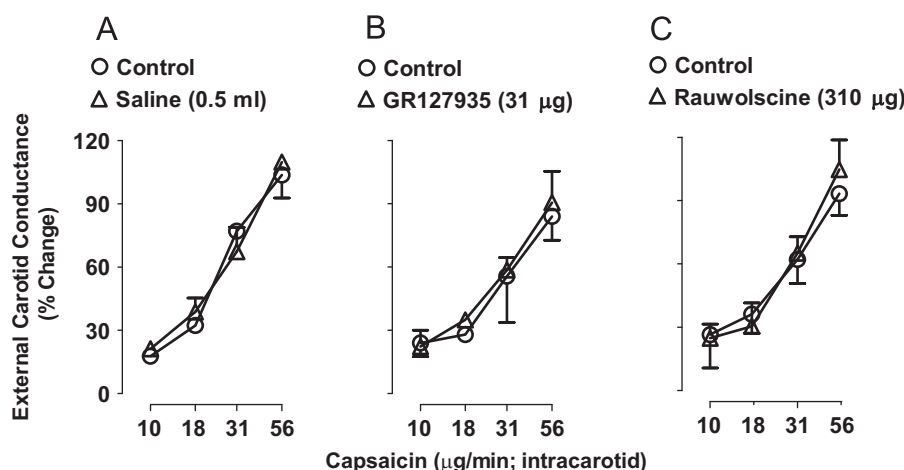


Fig. 3. Canine external carotid vasodilator responses produced by 1-min intracarotid infusions of capsaicin before (control responses) and after an intrathecal bolus injection of: (A) vehicle (0.5 ml physiological saline; $n=4$); (B) GR127935 (31 μg ; $n=4$); or (C) rauwolscline (310 μg ; $n=4$).

the external carotid conductance (see Fig. 2) without changes in mean blood pressure or heart rate (not shown), as previously reported (Muñoz-Islas et al., 2009; Villalón et al., 2012). Moreover, the vasodilator responses to capsaicin and α -CGRP were longer-lasting (between 5 and 25 min) than those to acetylcholine (between 1 and 5 min). These responses remained unchanged after intrathecal administration of propylene glycol 10% (Fig. 2A–C). In contrast, only the vasodilatation to capsaicin (Fig. 2D), but not that to α -CGRP (Fig. 2E) and acetylcholine (Fig. 2F), was significantly inhibited after intrathecal administration of 10 μg dihydroergotamine.

3.4. Effect of vehicle, GR127935 or rauwolscline on the vasodilatation to capsaicin or on dihydroergotamine-induced inhibition of the vasodilatation to capsaicin

As shown in Fig. 3, intrathecal administration of vehicle (0.5 ml saline; Fig. 3A), GR127935 (31 μg ; Fig. 3B) or rauwolscline (310 μg ; Fig. 3C) failed to significantly modify per se the vasodilator responses to capsaicin. Hence, the intrathecal doses of these compounds (and even lower doses; see below) are appropriate for identifying the pharmacological profile of dihydroergotamine-induced inhibition. On this basis, Fig. 4 shows that the inhibition by 10 μg dihydroergotamine on capsaicin-induced external carotid vasodilatation was: (i) unaffected by 0.5 ml saline (Fig. 4A), 10 μg GR127935 (Fig. 4B) or 100 μg rauwolscline (Fig. 4C); and (ii) abolished by 31 μg GR127935 (Fig. 4B) or 310 μg rauwolscline (Fig. 4C). These results led us to use another group of 6 dogs to investigate the effect produced by 10 μg GR127935 + 100 μg rauwolscline given intrathecally (which failed to revert, when given separately, the inhibition by 10 μg dihydroergotamine). This combination abolished the inhibition produced by the maximal dose (100 μg) of dihydroergotamine (Fig. 4D).

4. Discussion

4.1. General

Apart from the implications discussed below, our results show that direct intrathecal (C_1 – C_3) administration of dihydroergotamine inhibits the external carotid vasodilatation to capsaicin. This response seems to involve activation of serotonin 5-HT_{1B/1D} receptors (blocked by GR127935; Fig. 4B) and α_2 -adrenoceptors (blocked by rauwolscline; Fig. 4C). In keeping with this view, the

combination of both antagonists (at lower doses) abolished the inhibition by a higher dose (100 μg) of dihydroergotamine (Fig. 4D). These receptors most likely correlate with the serotonin 5-HT_{1B} subtype (as shown using intrathecal sumatriptan; Muñoz-Islas et al., 2009) and the $\alpha_{2A/2C}$ -adrenoceptor subtypes (as demonstrated using intrathecal B-HT 933; Villalón et al., 2012). For this reason, the effects of subtype-selective antagonists on dihydroergotamine-induced inhibition were not analyzed. Further, these findings do not seem to support the involvement of serotonin 5-HT_{1F} receptors (see also Table 1).

Regarding our experimental protocols, we did not measure sensory nerve activity directly, but the chemical (capsaicin-induced) neuropeptide release (most probably CGRP; Kapoor et al., 2003a) in the external carotid circulation could be estimated indirectly by measurement of the evoked external carotid vasodilator response.

4.2. Systemic and carotid haemodynamic changes by the different treatments

Intracarotid phenylephrine was continuously infused to enhance the vasodilator responses to capsaicin, α -CGRP and acetylcholine. This produced a sustained external carotid vasoconstriction (with no changes in blood pressure or heart rate; Table 2) due to activation of α_1 -adrenoceptors (Willems et al., 2001). Under these conditions, intrathecal administration of the vehicles (saline or propylene glycol), dihydroergotamine or the antagonists did not affect per se the baseline values of mean blood pressure, heart rate or external carotid conductance (see Tables 2 and 3). Besides, the external carotid vasodilator responses to capsaicin, α -CGRP and acetylcholine were reproducible as they remained unchanged after intrathecal vehicle (Fig. 2), as previously reported by our group (Muñoz-Islas et al., 2009; Villalón et al., 2012). Consequently, there were not time-dependent changes in the hemodynamic variables during the course of our experimental protocols.

The mechanisms involved in the above vasodilator responses seem to involve, as shown elsewhere: (i) capsaicin-induced CGRP release from perivascular nerves and activation of CGRP receptors (Kapoor et al., 2003a); (ii) CGRP receptors activated by exogenous α -CGRP (Kapoor et al., 2003b); and (iii) endothelial muscarinic receptors activated by acetylcholine (Villalón et al., 1993). As previously speculated (Villalón et al., 2012), capsaicin-induced external carotid vasodilatation may also be mediated by CGRP release from perivascular sensory nerves, but this hypothesis

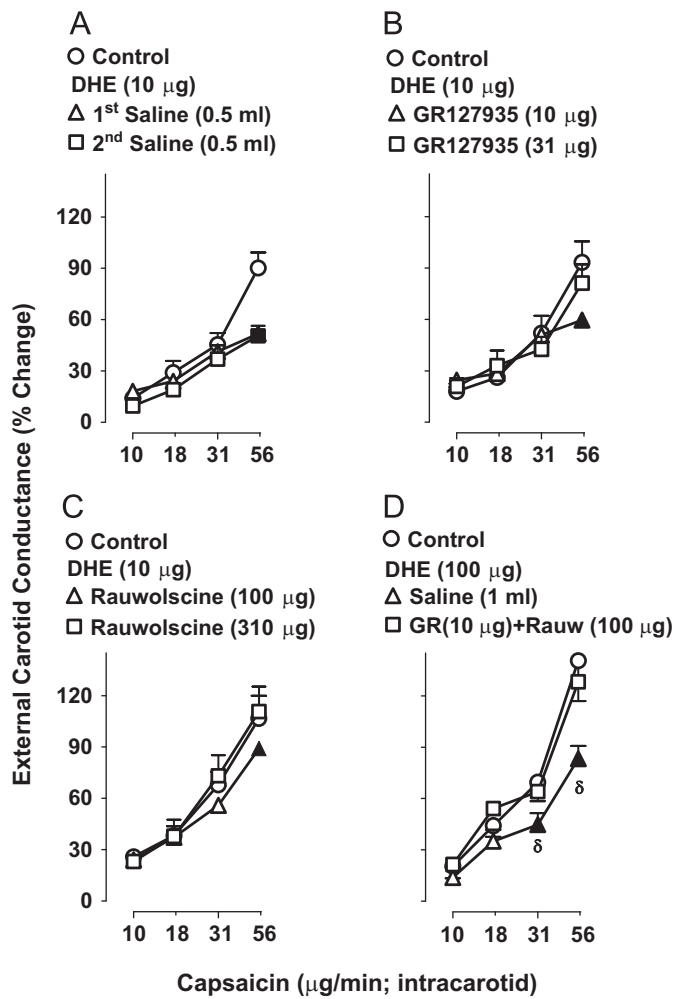


Fig. 4. Canine external carotid vasodilator responses by intracarotid infusions of capsaicin before (control responses) and after 10 µg dihydroergotamine (DHE, given intrathecally) in dogs treated intrathecally ($n=6$ each) with: (A) vehicle (0.5 ml saline given twice); (B) GR127935 (10 and 31 µg given cumulatively); and (C) rauwolscline (100 and 310 µg given cumulatively). In addition, (D) shows the inhibition of the responses to capsaicin after intrathecal DHE (100 µg) in dogs treated intrathecally with vehicle and the combination 10 µg GR127935+100 µg rauwolscline (10 µg GR+100 µg Rauw). For the sake of clarity, empty symbols depict either control responses (○) or non-significant responses (△) vs. control. Solid symbols (▲ ■) represent significantly different responses ($P < 0.05$) vs. control; δ, $P < 0.05$ vs. control or vs. the combination GR127935 + rauwolscline plus 100 µg DHE.

remains unproven. Clearly, further experiments will be required to confirm this assumption.

4.3. Specific inhibition by dihydroergotamine of the vasodilator responses to capsaicin

The fact that intrathecal dihydroergotamine inhibited the external carotid vasodilatation to capsaicin (particularly at high doses, where the central component of the vasodilator response may be more evident), but not that to α -CGRP or acetylcholine (Fig. 2) suggests that this response: (i) is highly specific; and (ii) involves a central (spinal) mechanism probably related to trigeminal inhibition rather than interactions with vascular CGRP receptors and/or muscarinic receptors. Moreover, the inhibition by dihydroergotamine is unrelated to its leakage into the circulation, since no significant changes in mean blood pressure, heart rate or external carotid conductance were observed (see Tables 2 and 3). The intrathecal doses of dihydroergotamine used

in this study (10–100 µg), in contrast with those of sumatriptan (1000 µg; Muñoz-Islas et al., 2009) and B-HT 933 (3100 µg; Villalón et al., 2012) seem adequate when considering its binding profile (Table 1). Indeed, this profile implies complex interactions with spinal serotonin 5-HT_{1B} receptor subtypes and $\alpha_{2A/2C}$ -adrenoceptor subtypes in our experimental model, in view that previous studies in dogs have excluded the role of spinal serotonin 5-HT_{1A}, 5-HT_{1D} and 5-HT_{1F} receptor subtypes (Muñoz-Islas et al., 2009) as well as α_{2B} -adrenoceptor subtypes (Villalón et al., 2012).

4.4. The role of spinal GR127935- and rauwolscline-sensitive receptors in the inhibition of the external carotid vasodilatation to capsaicin

To analyze the pharmacological profile of dihydroergotamine-induced inhibition of the carotid vasodilatation to capsaicin, it was firstly crucial to ascertain that the intrathecal administration of the antagonists GR127935 and rauwolscline (Table 1) were devoid of any effects per se on: (i) the vasodilator responses to capsaicin (Fig. 3); and (ii) mean blood pressure, heart rate and external carotid conductance (Table 3). Accordingly, any effect of these antagonists on the inhibition by dihydroergotamine can be interpreted in terms of a direct interaction of the antagonist with its corresponding spinal receptor.

Since the inhibition by 10 µg dihydroergotamine was abolished by 31 µg GR127935 or 310 µg rauwolscline, but unaffected by 10 µg GR127935 or 100 µg rauwolscline (Fig. 4), the simplest interpretation of these findings may suggest the role of serotonin 5-HT_{1B/1D} receptors and α_2 -adrenoceptors. However, this profile of blockade deserves further considerations in view of the complex binding properties of these compounds (Table 1).

Indeed, the lack of effect of 10 µg GR127935 or 100 µg rauwolscline is apparently unexpected if serotonin 5-HT_{1B/1D} receptors and α_2 -adrenoceptors are involved. Moreover, considering its binding properties (Table 1), dihydroergotamine may be simultaneously activating both these receptors and, consequently, the blockade of only one of these receptors (with either GR127935 or rauwolscline) may have been overshadowed by the activation of the other (non-blocked) receptor. Interestingly, higher doses of GR127935 (31 µg) or rauwolscline (310 µg) given separately may have produced a functional blockade of both receptors, a notion that is reinforced by their binding profile (Table 1) and the complete blockade of the inhibition by 10 µg dihydroergotamine (Fig. 4B and C). Consistent with this contention, the intrathecal administration of 10 µg GR127935+100 µg rauwolscline (which were devoid of any effect when given separately; see above) abolished the inhibition by a higher dose (100 µg) of dihydroergotamine (Fig. 4D). This might suggest (although does not prove) that the same result could have been obtained with the inhibition by 10 µg dihydroergotamine. Notwithstanding, we decided to make it sure that the complex interactions of dihydroergotamine in our study fall specifically within the bounds of serotonin 5-HT_{1B/1D} receptors and α_2 -adrenoceptors even when using a higher dose (100 µg).

4.5. A few minor experimental limitations and ethical considerations of our study

Admittedly, three experimental limitations of the present study are: (i) to ascertain in our canine experimental model whether either 10 µg GR127935 or 100 µg rauwolscline (in addition to completely blocking their respective receptors) is already starting to produce some degree of blockade of another (unrelated) receptor; (ii) to analyze the effect of 100 µg dihydroergotamine on the external carotid vasodilator responses to α -CGRP

and acetylcholine (i.e., producing no effect, as shown with 10 μ g dihydroergotamine); and (iii) to investigate the effect per se of the combination 10 μ g GR127935 + 100 μ g rauwolfscine on capsaicin-induced external carotid vasodilatation. However, it is noteworthy that these experiments involve the use of dogs, the number of which is strictly controlled (and requires compelling ethical reasons) by our ethics committee (CICUAL). In any case, the objective of the present study is accomplished (see Introduction), and the results to be obtained from these experiments would only reinforce previous findings with compounds given intrathecally (Muñoz-Islas et al., 2009; Villalón et al., 2012). Indeed, the present study shows no effect of: (i) 10 μ g dihydroergotamine on the external carotid vasodilator responses to α -CGRP and acetylcholine (Fig. 2); and (ii) either antagonist at a higher dose (Fig. 3), which abolished the inhibition by 10 μ g dihydroergotamine (Fig. 4).

4.6. Possible locus of the inhibition by dihydroergotamine: Potential clinical implications

Several lines of evidence have shown that activation of central serotonin 5-HT_{1B/1D} receptors produces trigeminal inhibition which, in turn, interferes with pain transmission (Buzzi et al., 1991; Knight et al., 2001; Nozaki et al., 1992). Most significantly, the fact that intrathecal (C₁–C₃) dihydroergotamine inhibited the external carotid vasodilatation to intracarotid capsaicin (Figs. 1 and 2) suggests the involvement of a spinal mechanism. This notion is consistent with other findings obtained from cat brain, in which specific binding sites for ³H-dihydroergotamine were identified in several regions including the dorsal horn of the cervical spinal cord, the medulla (associated with the nucleus of the tractus solitarius) and descending spinal trigeminal nucleus (Goadsby and Gundlach, 1991); these regions were also labeled after intravenous ³H-dihydroergotamine. Thus, these central regions labeled with ³H-dihydroergotamine may be involved in cranial pain transmission and, accordingly, may play a role in the antimigraine action of dihydroergotamine (particularly first-order trigeminal nucleus synapse in the brainstem and upper cervical cord). Admittedly, additional experiments, which fall beyond the scope of the present study, will be required to further investigate the role of these central regions in the inhibition produced by intrathecal dihydroergotamine on the extracranial vasodilatation to capsaicin.

Undeniably, dihydroergotamine has successfully been used for at least 60 years in aborting acute migraine attacks. Notwithstanding, more recently developed compounds such as the non-peptide CGRP receptor antagonists (gepants; Villalón and Olesen, 2009) and serotonin 5-HT_{1F} receptor agonists (lasmiditan; Ferrari et al., 2010) are: (i) effective in acute migraine therapy without producing cranial vasoconstriction; (ii) devoid of the vascular side effects typical of the triptans and ergots; and (ii) highly promissory and seem to represent an additional valuable option for acute migraine therapy.

5. Conclusion

Intrathecally dihydroergotamine inhibited the external carotid vasodilatation to capsaicin. This response is mainly mediated by spinal activation of serotonin 5-HT_{1B/1D} (probably 5-HT_{1B}) receptors and α_2 (probably $\alpha_{2A/2C}$)-adrenoceptors, most likely at the level of the trigeminocervical complex.

Acknowledgements

The authors would like to thank Mr. Arturo Contreras for his skilful technical assistance. We are also indebted to CONACyT (Mexico) for their financial support (grant 60789-M), and the pharmaceutical companies for their generous gifts (see Drugs Section).

References

- Akerman, S., Holland, P., Goadsby, P.J., 2011. Diencephalic and brainstem mechanisms in migraine. *Nat. Rev. Neurosci.* 12, 570–584.
- Buzzi, M.G., Carter, W.B., Shimizu, T., Heath, 3rd, H., Moskowitz, M.A., 1991. Dihydroergotamine and sumatriptan attenuate levels of CGRP in plasma in rat superior sagittal sinus during electrical stimulation of the trigeminal ganglion. *Neuropharmacology* 30, 1193–1200.
- Caterina, M.J., Schumacher, M.A., Tominaga, M., Rosen, T.A., Levine, J.D., Julius, D., 1997. The capsaicin receptor: a heat-activated ion channel in the pain pathway. *Nature* 389, 816–824.
- Dodick, D.W., 2008. Examining the essence of migraine—is it the blood vessel or the brain? A debate. *Headache* 48, 661–667.
- Dux, M., Sántha, P., Jancsó, G., 2003. Capsaicin-sensitive neurogenic sensory vasodilatation in the dura mater of the rat. *J. Physiol.* 552, 859–867.
- Ferrari, M.D., Farkkila, M., Reuter, U., Pilgrim, A., Davis, C., Krauss, M., et al., 2010. Acute treatment of migraine with the selective 5-HT_{1F} receptor agonist lasmiditan—a randomised proof-of-concept trial. *Cephalalgia* 30, 1170–1178.
- Goadsby, P.J., Gundlach, A.L., 1991. Localization of ³H-dihydroergotamine-binding sites in the cat central nervous system: relevance to migraine. *Ann. Neurol.* 29, 91–94.
- Graham, J.R., Wolff, H.G., 1938. Mechanisms of migraine headache and action of ergotamine tartrate. *Arch. Neurol. Psychiat.* 39, 737–763.
- Gupta, S., Villalón, C.M., 2010. The relevance of preclinical research models for the development of antimigraine drugs: focus on 5-HT_{1B/1D} and CGRP receptors. *Pharmacol. Ther.* 128, 170–190.
- Jasper, J.R., Lesnick, J.D., Chang, L.K., Yamanishi, S.S., Chang, T.K., Hsu, S.A., Daunt, D.A., Bonhaus, L.R., Eglen, R.M., 1998. Ligand efficacy and potency at recombinant α_2 adrenergic receptors: agonist-mediated [³⁵S]GTPgammaS binding. *Biochem. Pharmacol.* 55, 1035–1043.
- Jiménez-Mena, L.R., Gupta, S., Muñoz-Islas, E., Lozano-Cuenca, J., Sánchez-López, A., Centurión, D., Mehrotra, S., MaassenVanDenBrink, A., Villalón, C.M., 2006. Clonidine inhibits the canine external carotid vasodilatation to capsaicin by $\alpha_{2A/2C}$ -adrenergic receptors. *Eur. J. Pharmacol.* 543, 68–76.
- Kapoor, K., Arulmani, U., Heiligers, J.P.C., Garrelds, I.M., Willems, E.W., Doods, H., Villalón, C.M., Saxena, P.R., 2003a. Effects of the CGRP receptor antagonist BIBN4096BS on capsaicin-induced carotid haemodynamic changes in anaesthetised pigs. *Br. J. Pharmacol.* 140, 329–338.
- Kapoor, K., Arulmani, U., Heiligers, J.P.C., Willems, E.W., Doods, H., Villalón, C.M., Saxena, P.R., 2003b. Effects of BIBN4096BS on cardiac output distribution and on CGRP-induced carotid haemodynamic responses in the pig. *Eur. J. Pharmacol.* 475, 69–77.
- Kayser, V., Aubel, B., Hamon, M., Bourgoin, S., 2002. The antimigraine 5-HT_{1B/1D} receptor agonists, sumatriptan, zolmitriptan and dihydroergotamine, attenuate pain-related behaviour in a rat model of trigeminal neuropathic pain. *Br. J. Pharmacol.* 137, 1287–1297.
- Kim, K.S., Yoo, H.Y., Park, K.S., Kim, J.K., Zhang, Y.H., Kim, S.J., 2012. Differential effects of acute hypoxia on the activation of TRPV1 by capsaicin and acidic pH. *J. Physiol. Sci.* 62, 93–103.
- Kleinman, L.L., Radford, E.P., 1964. Ventilation standards for small mammals. *J. Appl. Physiol.* 19, 360–362.
- Knight, Y.E., Edvinsson, L., Goadsby, P.J., 2001. 4991W93 inhibits release of calcitonin gene-related peptide in the cat but only at doses with 5-HT_{1B/1D} receptor agonist activity? *Neuropharmacology* 40, 520–525.
- Leyssen, J.E., Gommeren, W., Heylen, L., Luyten, W.H., Van de Weyer, I., Vanhorenacker, P., Haegeman, G., Schotte, A., Van Gompel, P., Wouters, R., Lesage, A.S., 1996. Alniditan, a new 5-hydroxytryptamine_{1D} agonist and migraine-abortive agent: ligand-binding properties of human 5-hydroxytryptamine_{1D} α -human 5-hydroxytryptamine_{1D} β - and calf 5-hydroxytryptamine_{1D} receptors investigated with [³H]5-hydroxytryptamine and [³H]alniditan. *Mol. Pharmacol.* 50, 1567–1580.
- Link, A.S., Kuris, A., Edvinsson, L., 2008. Treatment of migraine attacks based on the interaction with the trigemino-cerebrovascular system. *J. Headache Pain* 9, 5–12.
- Moskowitz, M.A., Reinhard Jr., J.F., Romero, J., Melamed, E., Pettibone, D.J., 1979. Neurotransmitters and the fifth cranial nerve: is there a relation to the headache phase of migraine? *Lancet* 2, 883–885.
- Muñoz-Islas, E., Gupta, S., Jiménez-Mena, L.R., Lozano-Cuenca, J., Sánchez-López, A., Centurión, D., Mehrotra, S., MaassenVanDenBrink, A., Villalón, C.M., 2006. Donatriptan, but not sumatriptan, inhibits capsaicin-induced canine external carotid vasodilatation via 5-HT_{1B} rather than 5-HT_{1D} receptors. *Br. J. Pharmacol.* 149, 82–91.
- Muñoz-Islas, E., Lozano-Cuenca, J., González-Hernández, A., Ramírez-Rosas, M.B., Sánchez-López, A., Centurión, D., MaassenVanDenBrink, A., Villalón, C.M.,

2009. Spinal sumatriptan inhibits capsaicin-induced canine external carotid vasodilatation via 5-HT_{1B} rather than 5-HT_{1D} receptors. *Eur. J. Pharmacol.* 615, 133–138.
- Nozaki, K., Moskowitz, M.A., Boccalini, P., 1992. CP-93,129, sumatriptan, dihydroergotamine block c-fos expression within rat trigeminal nucleus caudalis caused by chemical stimulation of the meninges. *Br. J. Pharmacol.* 106, 409–415.
- Olesen, J., Burstein, R., Ashina, M., Tfelt-Hansen, P., 2009. Origin of pain in migraine: evidence for peripheral sensitisation. *Lancet Neurol.* 8, 679–690.
- Pauwels, P.J., 1996. Pharmacological properties of a putative 5-HT_{1B/1D} receptor antagonist GR127935. *CNS Drug Rev.* 2, 415–428.
- Price, G.W., Burton, M.J., Collin, L.J., Duckworth, M., Gaster, L., Gothert, M., Jones, B.J., Roberts, C., Watson, J.M., Middlemiss, D.N., 1997. SB-216641 and BRL-15572-compounds to pharmacologically discriminate h5-HT_{1B} and h5-HT_{1D} receptors. *Naunyn-Schmiedeberg's Arch. Pharmacol.* 356, 312–320.
- Saito, K., Markowitz, S., Moskowitz, M.A., 1988. Ergot alkaloids block neurogenic extravasation in dura mater: proposed action in vascular headaches. *Ann. Neurol.* 24, 732–737.
- Saper, J.R., Silberstein, S., 2006. Pharmacology of dihydroergotamine and evidence for efficacy and safety in migraine. *Headache* 46, S171–S181.
- Saxena, P.R., de Vlaam-Schluter, G.M., 1974. Role of some biogenic substances in migraine and relevant mechanism in antimigraine action of ergotamine—studies in an experimental model for migraine. *Headache* 13, 142–163.
- Shevel, E., 2011a. The extracranial vascular theory of migraine—a great story confirmed by the facts. *Headache* 51, 409–417.
- Shevel, E., 2011b. The extracranial vascular theory of migraine: an artificial controversy. *J. Neural Transm.* 118, 525–530.
- Uhlén, S., Porter, A.C., Neubig, R.R., 1994. The novel alpha-2 adrenergic radioligand [³H]-MK912 is alpha-2C selective among human alpha-2A, alpha-2B and alpha-2C adrenergic receptors. *J. Pharmacol. Exp. Ther.* 271, 1558–1565.
- Villalón, C.M., Centurión, D., Valdivia, L.F., De Vries, P., Saxena, P.R., 2003. Migraine: pathophysiology, pharmacology, treatment and future trends. *Curr. Vasc. Pharmacol.* 1, 71–84.
- Villalón, C.M., Centurión, D., Willems, E.W., Arulmani, U., Saxena, P.R., Valdivia, L.F., 2004. 5-HT_{1B} receptors and alpha_{2A/2C}-adrenergic receptors mediate external carotid vasoconstriction to dihydroergotamine. *Eur. J. Pharmacol.* 484, 287–290.
- Villalón, C.M., De Vries, P., Rabelo, G., Centurión, D., Sánchez-López, A., Saxena, P.R., 1999. Canine external carotid vasoconstriction to methysergide, ergotamine and dihydroergotamine: role of 5-HT_{1B/1D} receptors and α_2 -adrenergic receptors. *Br. J. Pharmacol.* 126, 585–594.
- Villalón, C.M., Galicia-Carreón, J., González-Hernández, A., Marichal-Cancino, B.A., Manrique-Maldonado, G., Centurión, D., 2012. Pharmacological evidence that spinal α_{2C} - and, to a lesser extent, α_{2A} -adrenoceptors inhibit capsaicin-induced vasodilatation in the canine external carotid circulation. *Eur. J. Pharmacol.* 683, 204–210.
- Villalón, C.M., Olesen, J., 2009. The role of CGRP in the pathophysiology of migraine and efficacy of CGRP receptor antagonists as acute antimigraine drugs. *Pharmacol. Ther.* 124, 309–323.
- Villalón, C.M., Ramírez-San Juan, E., Castillo, C., López-Muñoz, F.J., Terrón, J.A., 1995. Pharmacological profile of the receptors that mediate external carotid vasoconstriction by 5-HT in vagosympathectomized dogs. *Br. J. Pharmacol.* 116, 2778–2784.
- Villalón, C.M., Terrón, J.A., Hong, E., 1993. Role of 5-HT₁-like receptors in the increase in external carotid blood flow induced by 5-hydroxytryptamine in the dog. *Eur. J. Pharmacol.* 240, 9–20.
- Wang, Y., Aun, R., Tse, L.S.F., 1998. Brain uptake of dihydroergotamine after intravenous and nasal administration in the rat. *Biopharm. Drug Dispos.* 19, 571–575.
- Willems, E.W., Valdivia, L.F., Saxena, P.R., Villalón, C.M., 2001. The role of several α_1 - and α_2 -adrenergic receptor subtypes mediating vasoconstriction in the canine external carotid circulation. *Br. J. Pharmacol.* 132, 1292–1298.
- Yaksh, T.L., Horais, K.A., Tozier, N., Rathbun, M., Richter, P., Rossi, S., Grafe, M., Tong, C., Meschter, C., Cline, J.M., Eisenach, J., 2004. Intrathecal ketorolac in dogs and rats. *Toxicol. Sci.* 80, 322–334.